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Heredity

Mendelian Inheritance



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Heredity

- Genetics is the study of heredity
- Heredity is the passing of characters (traits) from the parents to the offspring
- Both the father and mother are contribute to the traits in the offspring
- Gregor Mendel explained how heredity takes place
- These are **Mendel's Laws of Inheritance**

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Functional Terms

- **Fertilization** - The uniting of male (sperm) & female (egg) gametes (sex cells)
- **Gene -** Located on the chromosome, they control how an organism develops
  - A particular segment of DNA which determines the heredity of a particular trait
- **Trait –** The characters that are passed on from parents to the offspring
- **Locus –** is the spot each gene has on chromosome

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Functional Trait

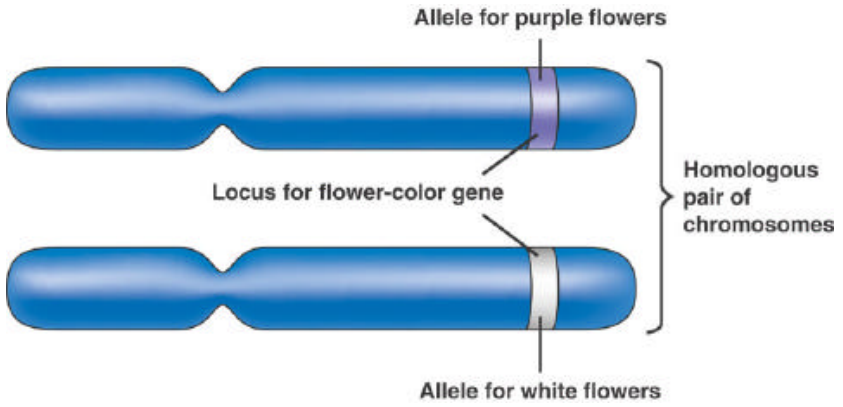
- **Alleles –** Different forms of same gene - each organism has two alleles each trait (exceptions)
- **Homozygous –** If both the alleles for a trait are similar (TT)
- **Heterozygous –** If both the alleles for a trait are different (Tt)
- **Phenotype –** The way an organism looks – Tall or Short
- **Genotype –** The genetic make up or genetic combination or constitution of an individual – TT, Rr, Aa etc.

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Meiosis & Inheritance

- Meiosis is the basis for heredity
- Chromosomes carry genes and each locus on a chromosome has alternative versions of the gene – allele
- We inherit one allele from each parent
- An allele can be dominant or recessive

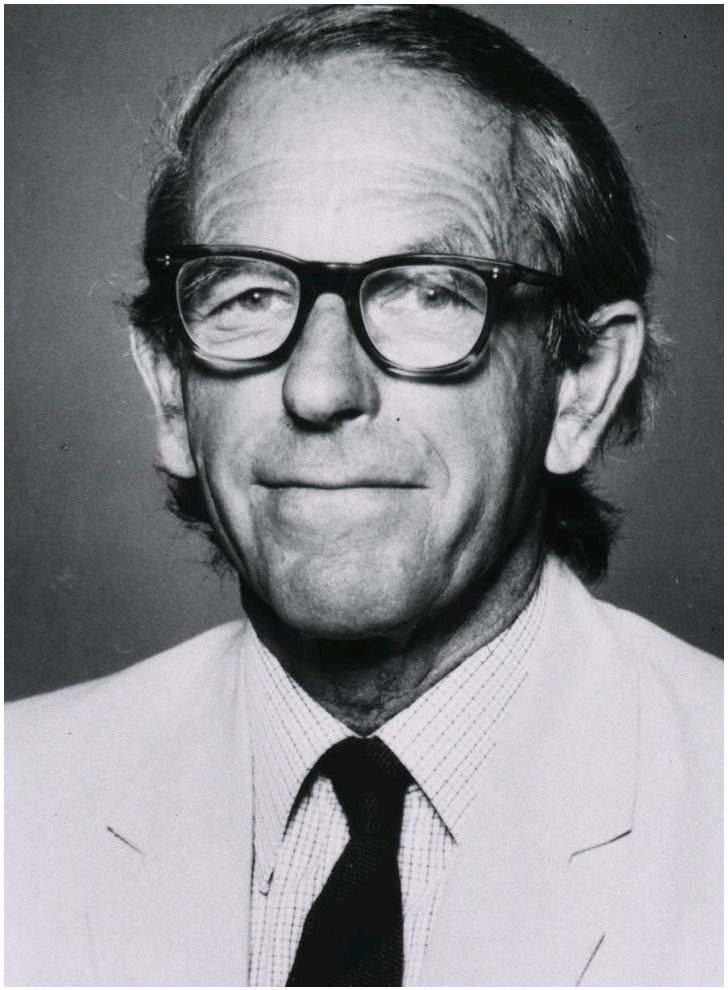


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Gregor Johann Mendel

- Austrian Monk, Mendel's experiments and observations led to **Principles of Genetics**
- Experiments on 29,000 garden pea (Pisum sativum) between 1856 & 1863
- His generalizations on segregation and independent assortment were not widely accepted
- He died in 1884 and not until 1900s that his work was rediscovered by De Vries and Carl Correns



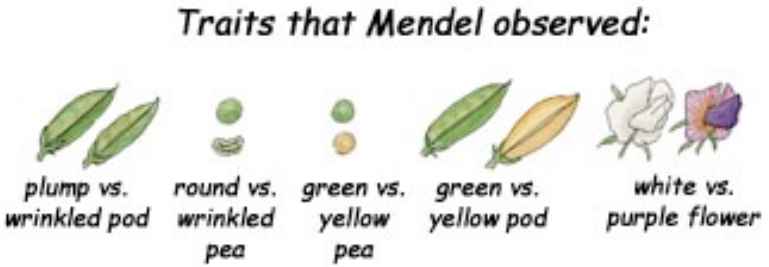
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Mendel's Procedures

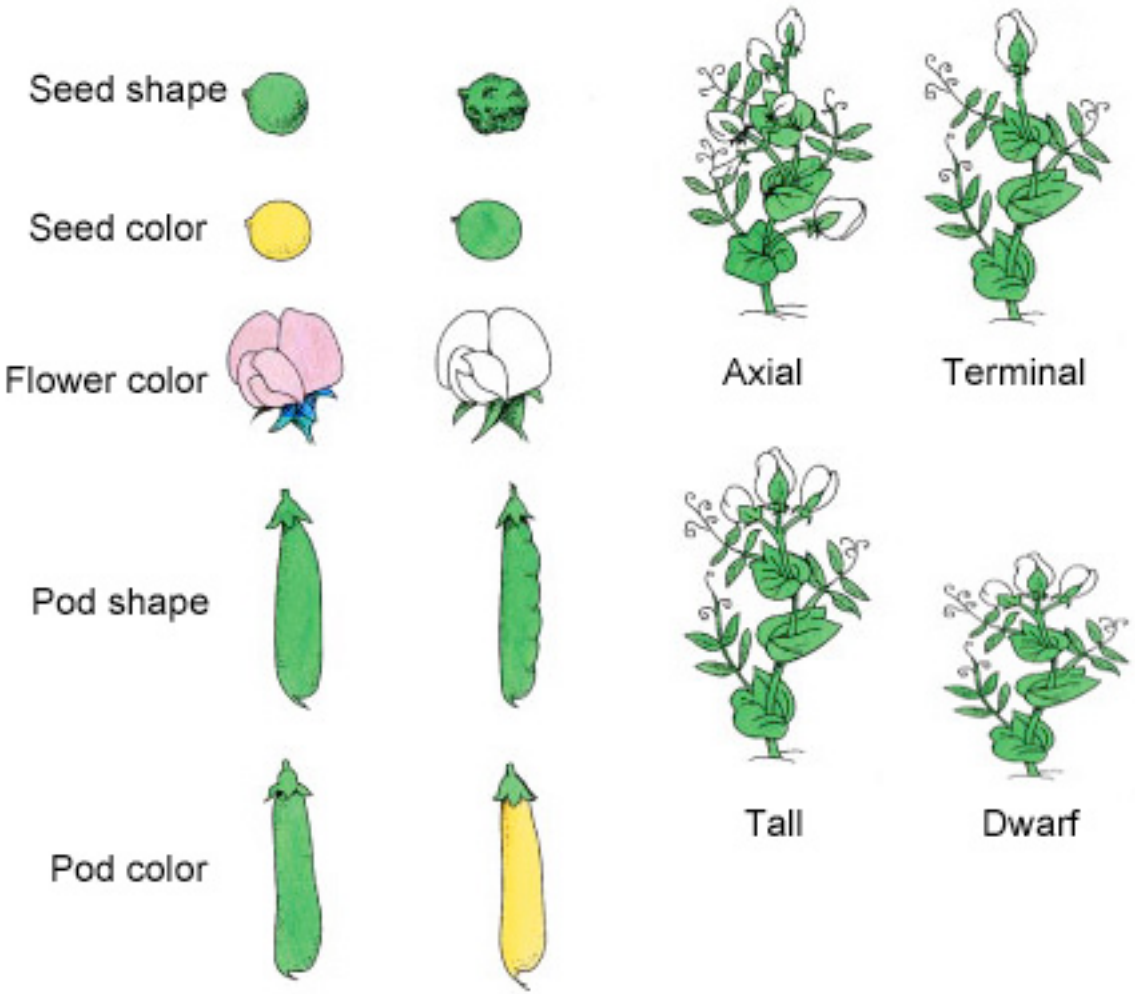
- He chose Pisum sativum (Garden Pea) because
  - Sexually reproducing
  - They have two gametes – male and female,
  - Their traits are easy to isolate and have wide variation
  - Convenience in handling





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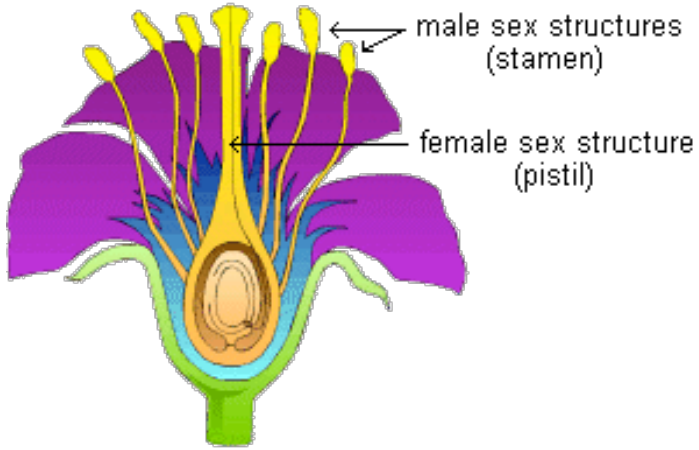


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Mendel's Procedures

- They have both male and female sexual reproductive organs and as a result they can self –pollinate themselves or cross-pollinate other plants
- Mendel was able to selectively cross pollinate *purebred* plants with particular traits and observe the outcomes over many generations
- These are the bases for his conclusions about nature of genetic inheritance



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Rule of Unit Factors

- An Organism has two factors that control each of its traits
- These factors, as we know them today, are genes and the variants of the gene are alleles
- For a particular trait, an individual inherits one such factor (gene) from each parent

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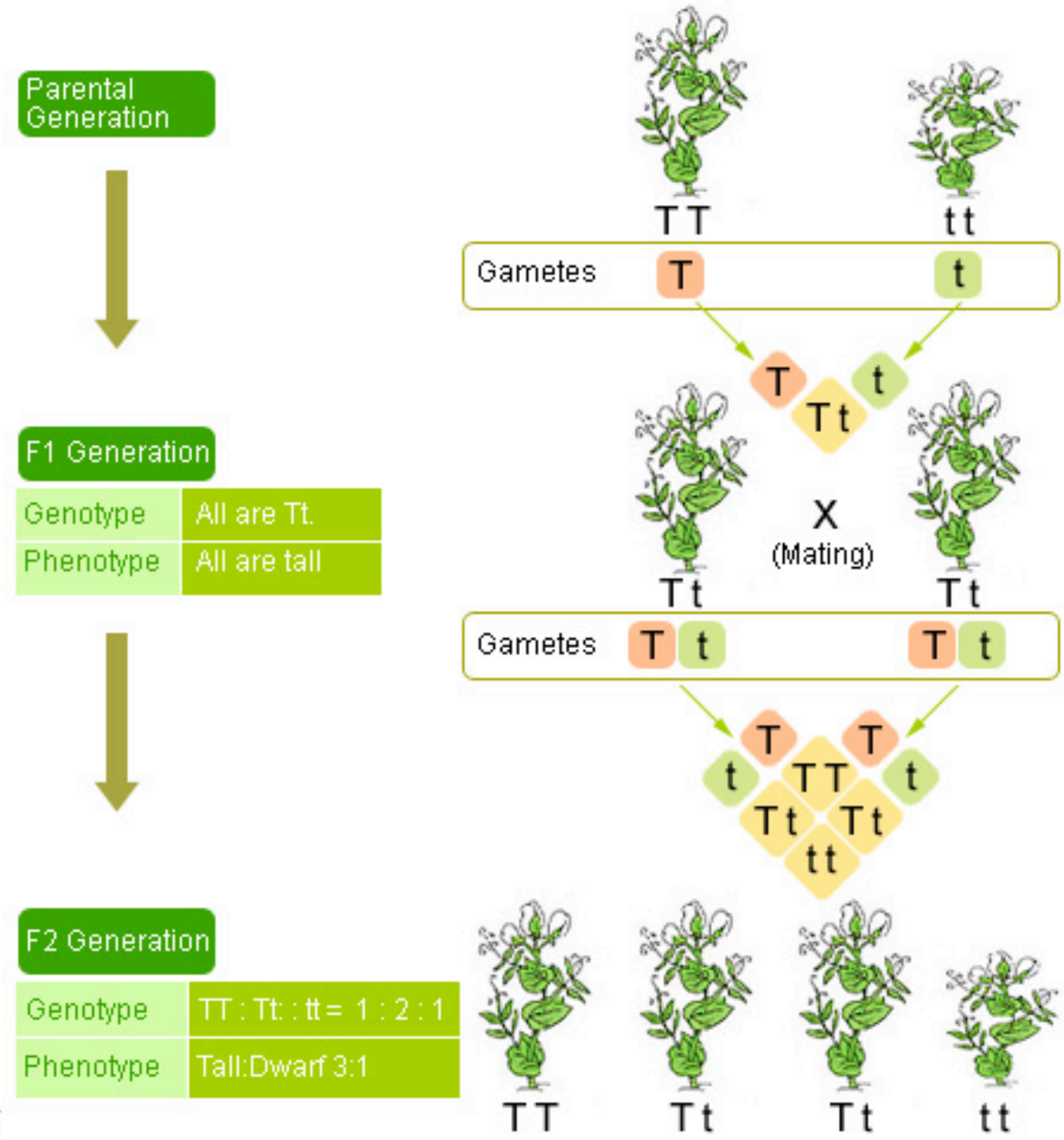
Rule of Dominance

- **In a cross between two parents differing in a character (say one parent is tall and the other is dwarf), in the first generation only one character appears**
- The character that appears in the first generation is called the **Dominant Character**
- The character that disappears in the first generation and reappears in the next generation is called the **Recessive Character**
- Alleles of dominant character are represented in **uppercase** and the recessive in **lowercase**

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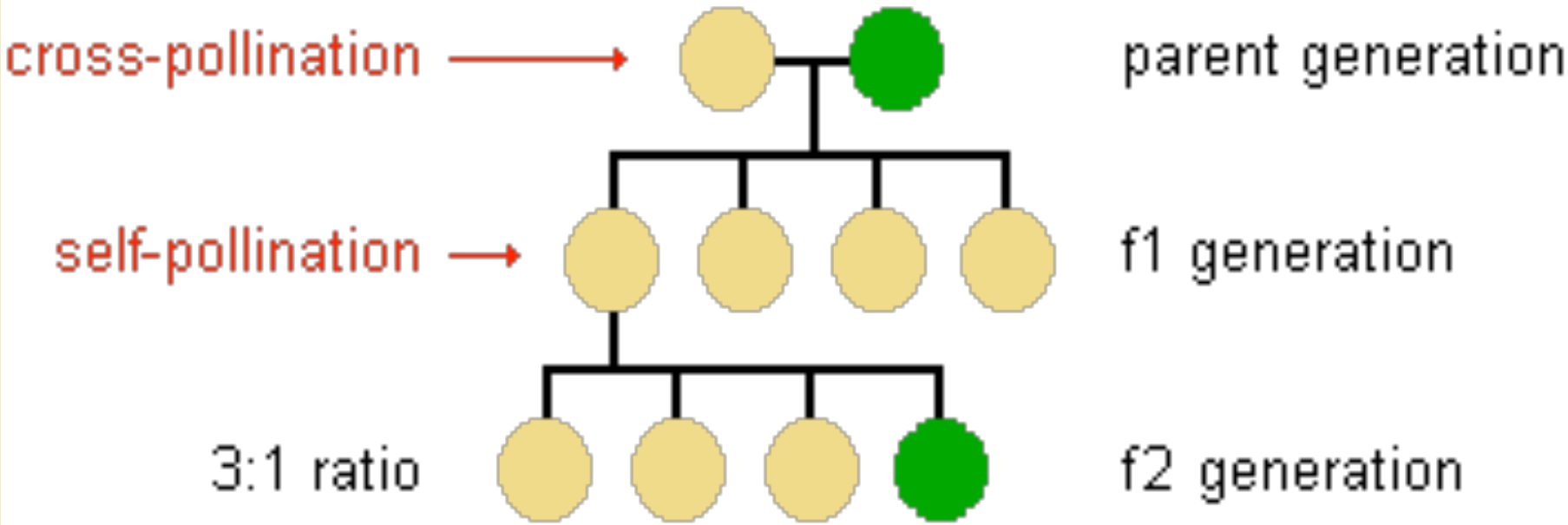
Mechanism of Dominance



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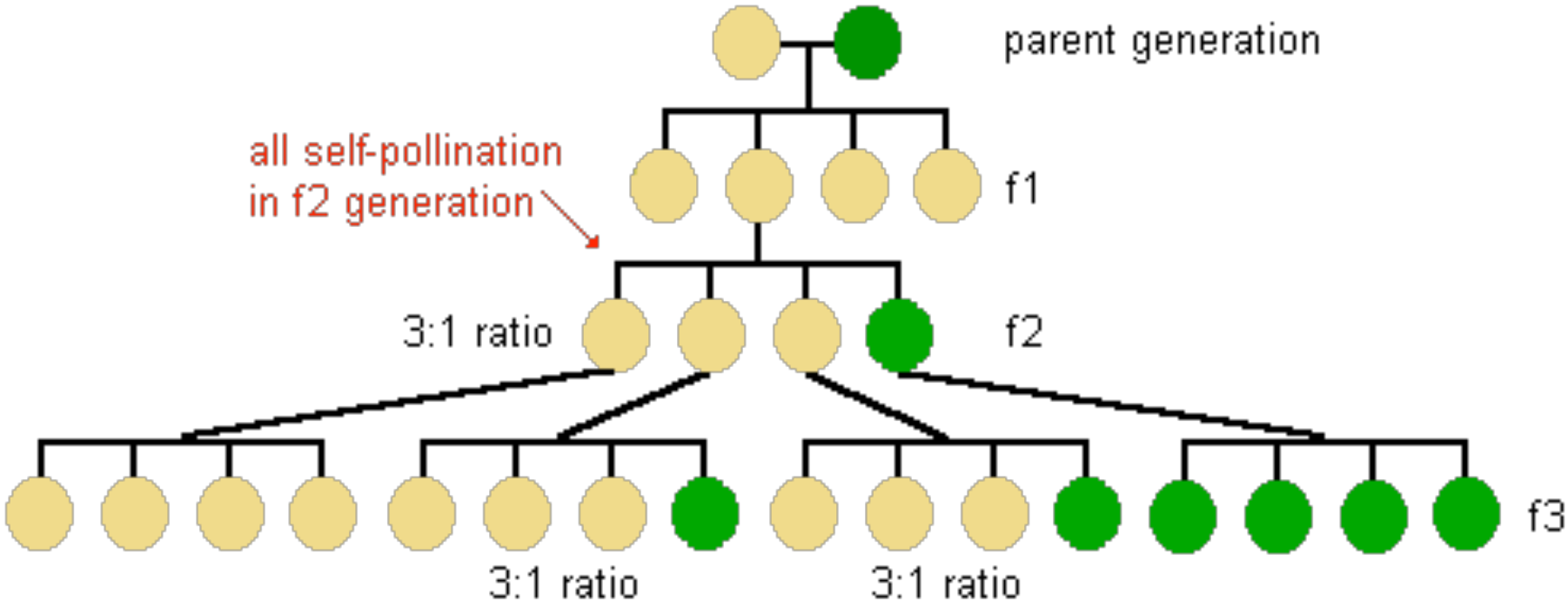
Mechanism of Dominance



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Mechanism of Dominance



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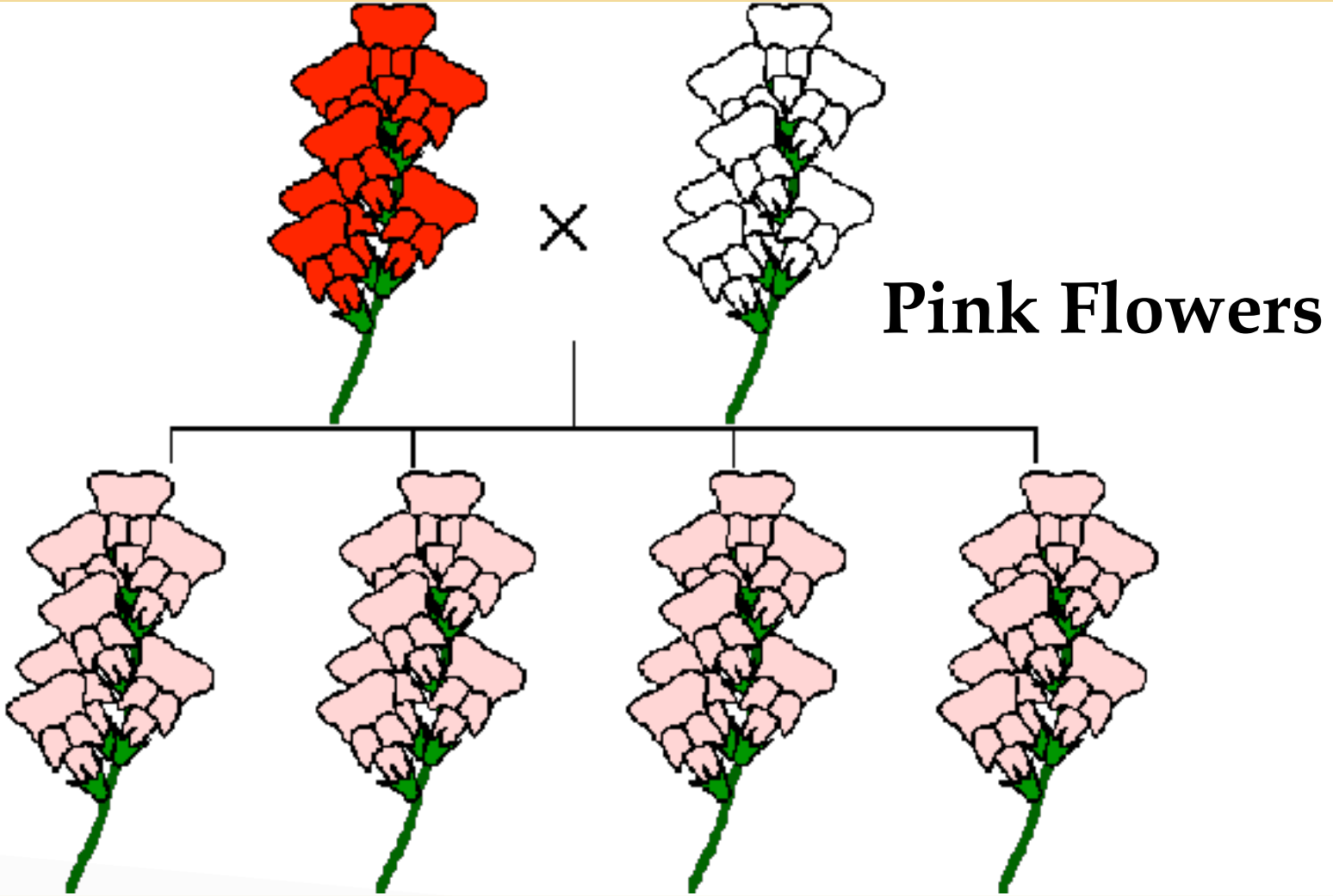
Exception to Dominance

- Exceptions exist for dominance
- **Incomplete Dominance** - where the dominant trait does not completely express itself in heterozygous condition – Ex. Rose
- **Co-dominance –** where both the alleles (dominant and recessive) are "equally" expressive in heterozygous condition

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Incomplete Dominance



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Co-dominance







Roan Cattle



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Law of Segregation

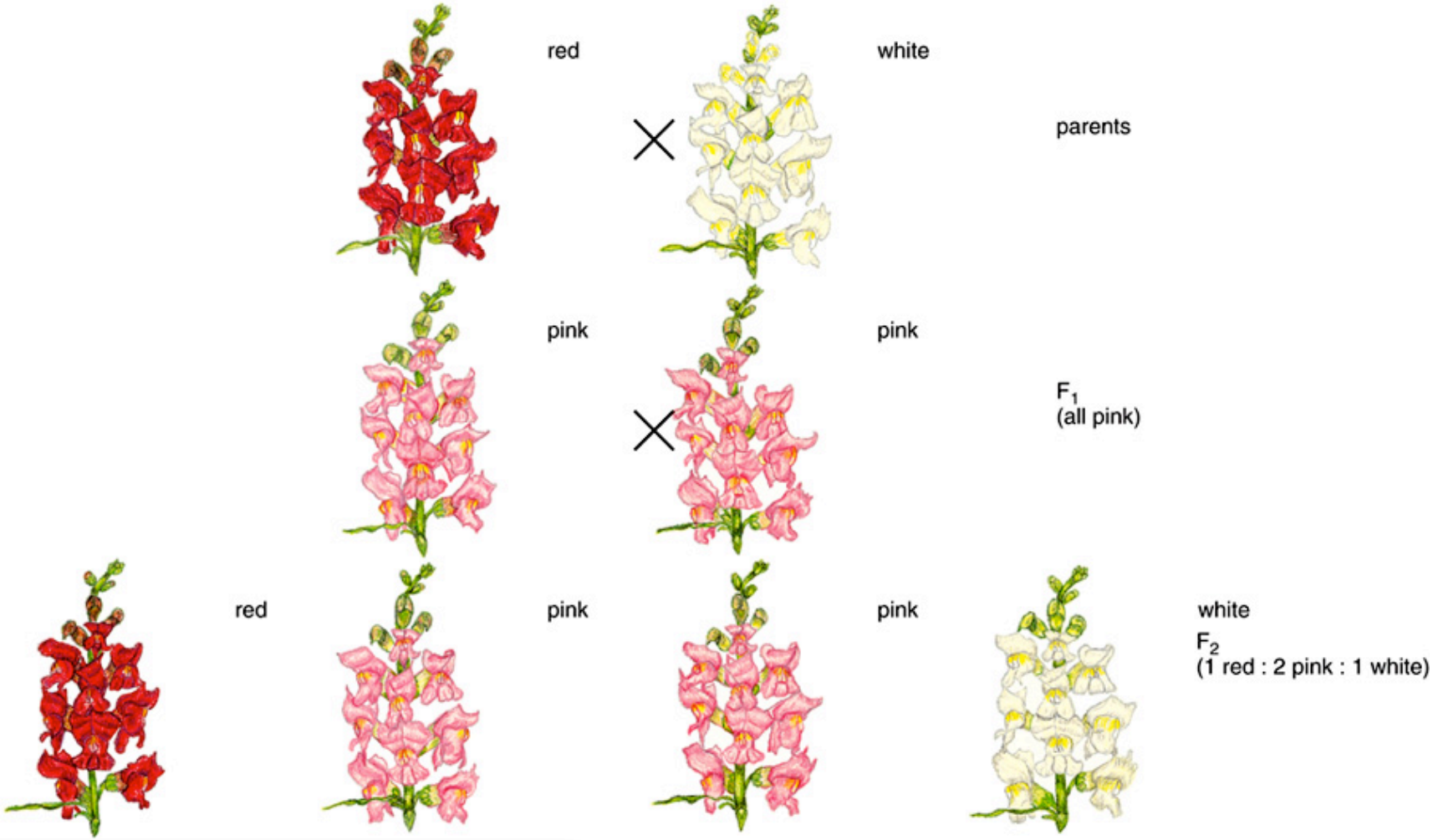
- The hybrids of F1 generation have alleles for two contrasting characters - dominant and recessive nature (Tt)
- **Though they remain together, these characters / alleles will not contaminate each other and segregate or separate during gametogenesis**
- **A parent randomly passes only one allele for each trait to the offspring**
- During gametogenesis, each gamete receives only one character – either dominant or recessive (reduction division)
- Ex. Segregation in offspring showing incomplete dominance

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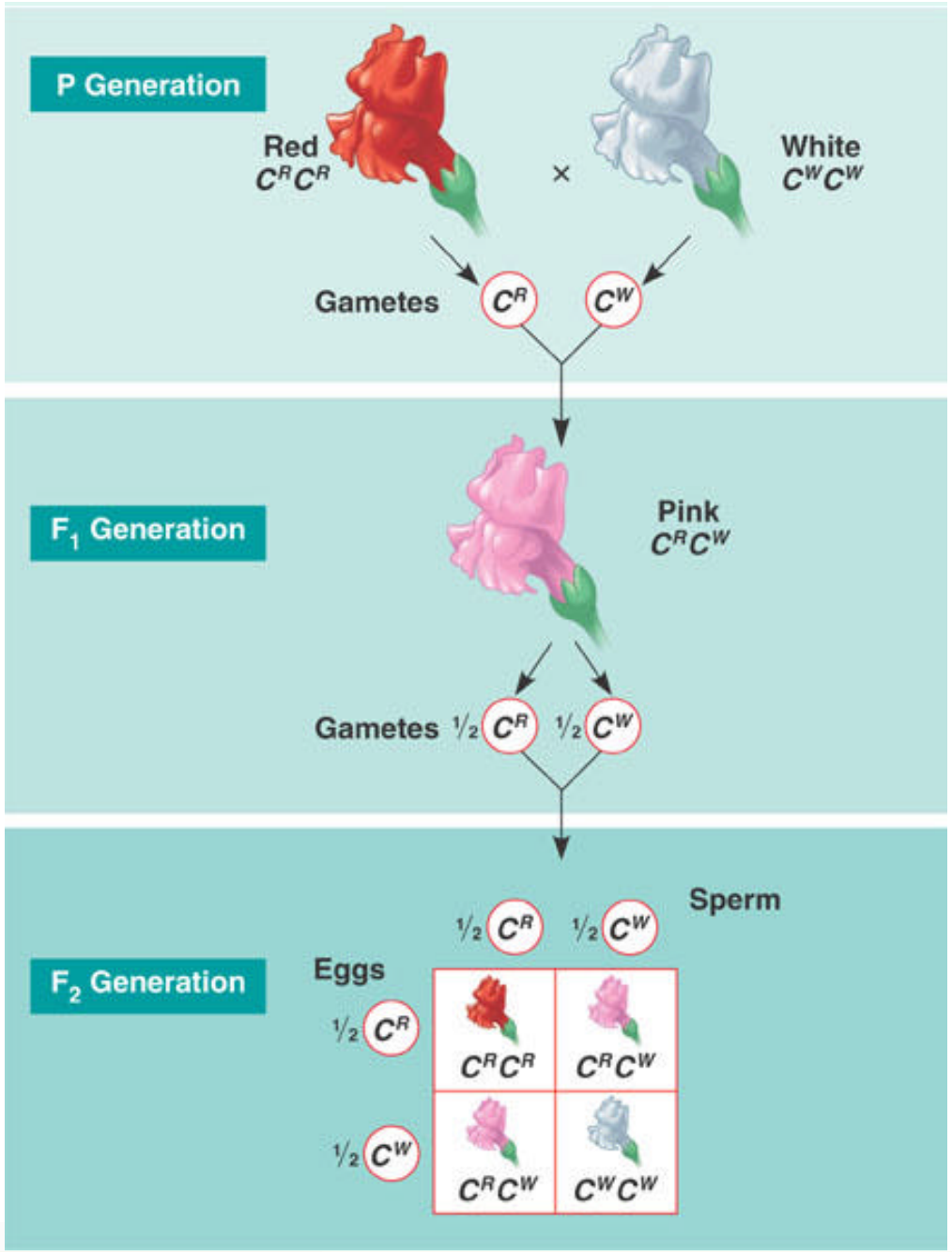
Mechanism of Segregation

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Monohybrid Vs Dihybrid Cross

- For dominance and segregation, Mendel considered Monohybrid cross but for Law of Independent Assortment, a Dihybrid Cross is considered
- **Monohybrid Cross –** Crossing two parents who differ in only one trait – Ex. Stature of the Plant – Tall or Dwarf (TT or tt)
- **Dihybrid Cross –** Crossing two parents who differ in two traits – Ex. Color of the Seed and Shape of the Seed (YYRR or yyrr)

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Independent Assortment

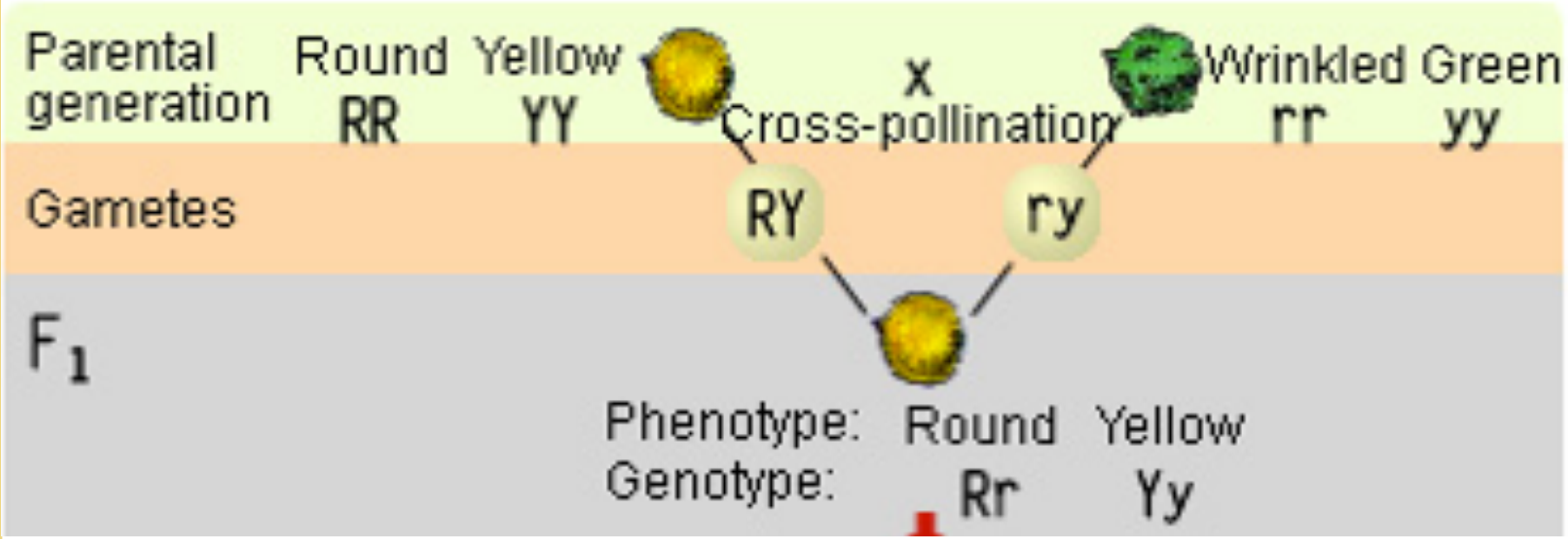
- **When parents differing from each other in two or more traits cross, then the inheritance of one trait is independent of the other trait**
- **The genes for different traits (on unlinked loci) are inherited independently of each other**
- Mendel used Dihybrid Cross to consider independent assortment
- Seed Color – Yellow (Dom) or Green (Rec) – Y or y
- Seed Shape – Round (Dom) or Wrinkled (Rec) – R or r

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Mechanism of I.A.

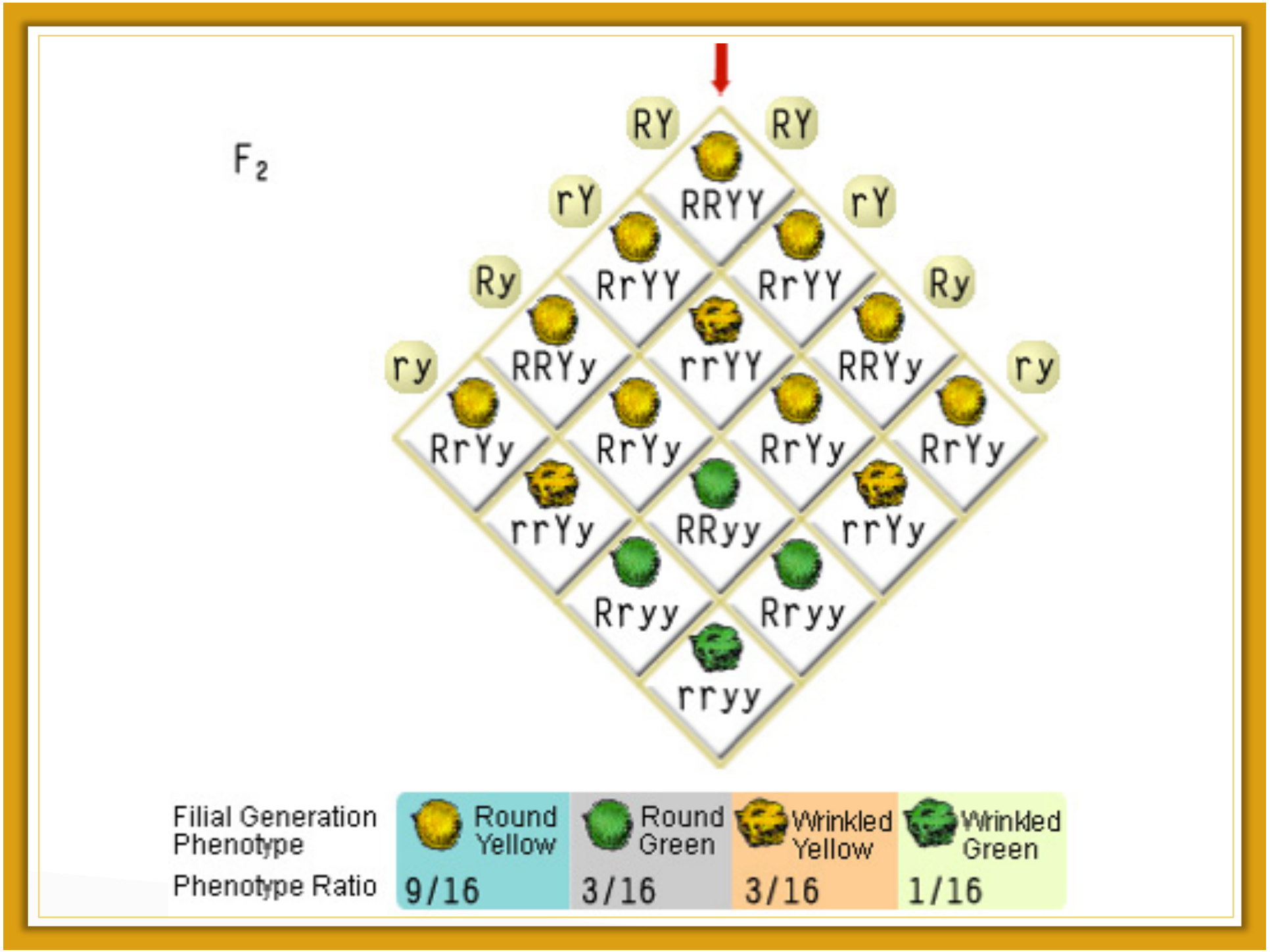
RRYY X rryy



RrYy F1 Self Fertilization – RrYy X RrYy

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Mendelian Inheritance in Humans

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Mendelian Traits in Man

- Ability to taste phenylthiocarbamide (dominant)
- ACHOO syndrome (dominant) (Photo sneeze reflex)
- Albinism (recessive)
- Cleft chin (dominant)
- Hitchhiker's thumb (recessive)



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Mendelian Traits in Man

- Free (dominant) or attached (recessive) earlobes
- Wet (dominant) or dry (recessive) earwax
- Face freckles (dominant)
- Huntingtons Disease (neurodegenerative) (dominant)
- Widow's peak (dominant)



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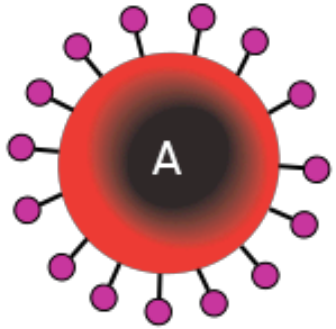
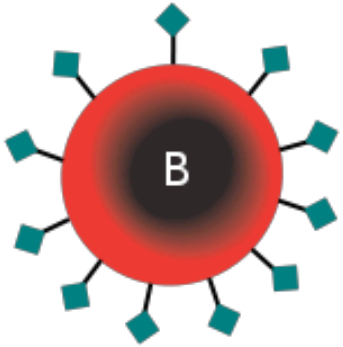
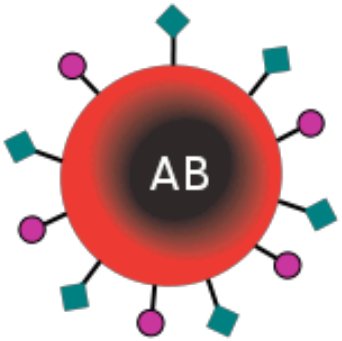
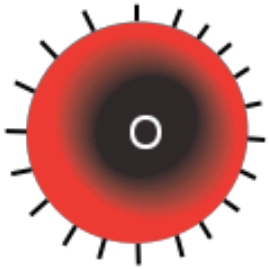
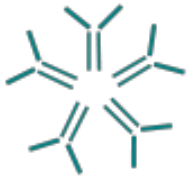
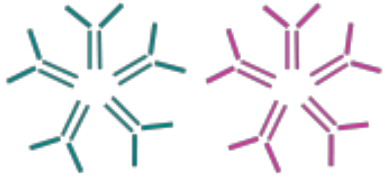
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Genetics of Blood Groups

- There are more than 30 different types of blood group systems
- ABO, Rh are popular
- But there are others like MNS, Kell, Lewis, Duffy etc.
- Blood groups are classified based on the presence or absence of specific antigens on the surface of RBCs
- Some blood groups have known Mendelian inheritance like ABO

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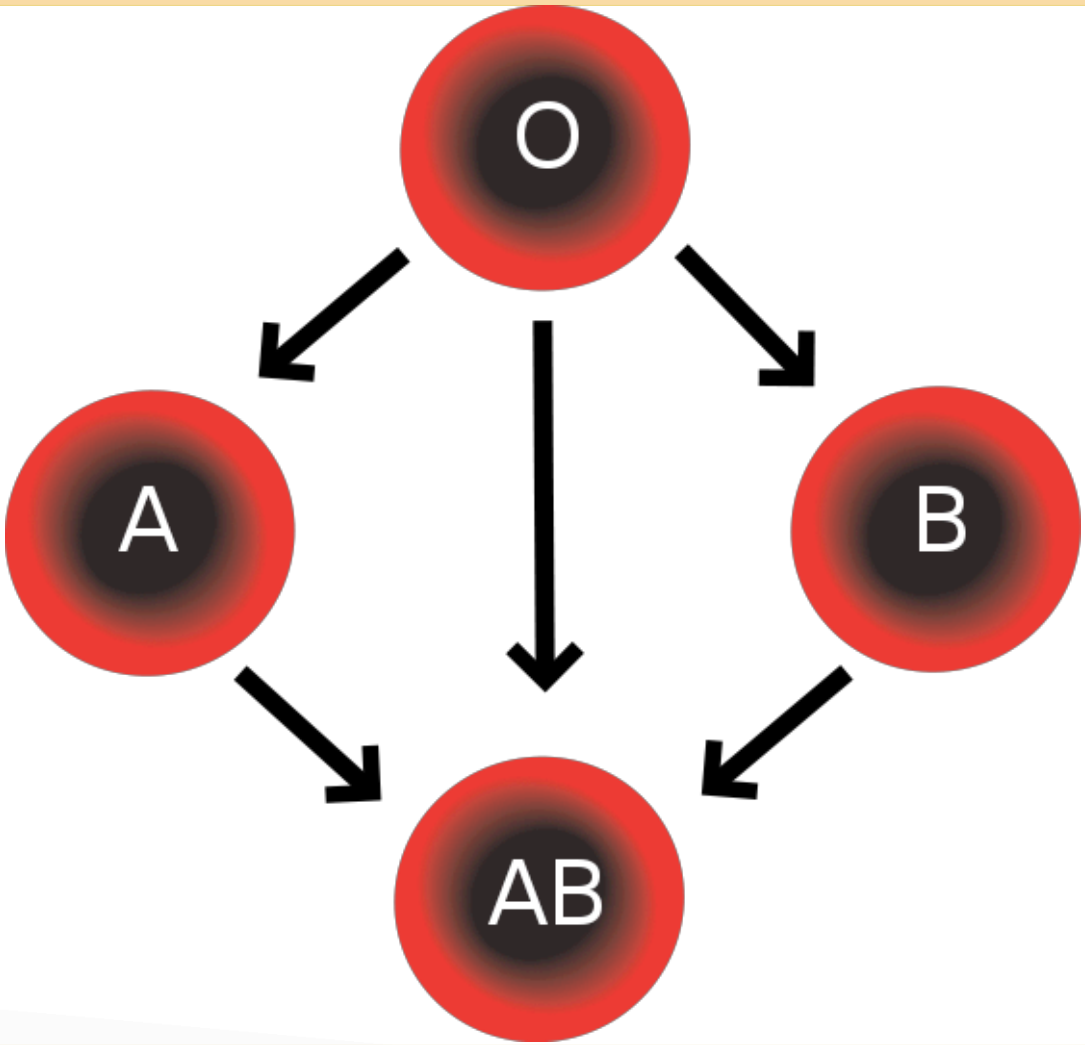
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	Group A	Group B	Group AB	Group O
Red blood cell type				
Antibodies present	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens present	 A antigen	 B antigen	 A and B antigens	None

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ABO Compatibility

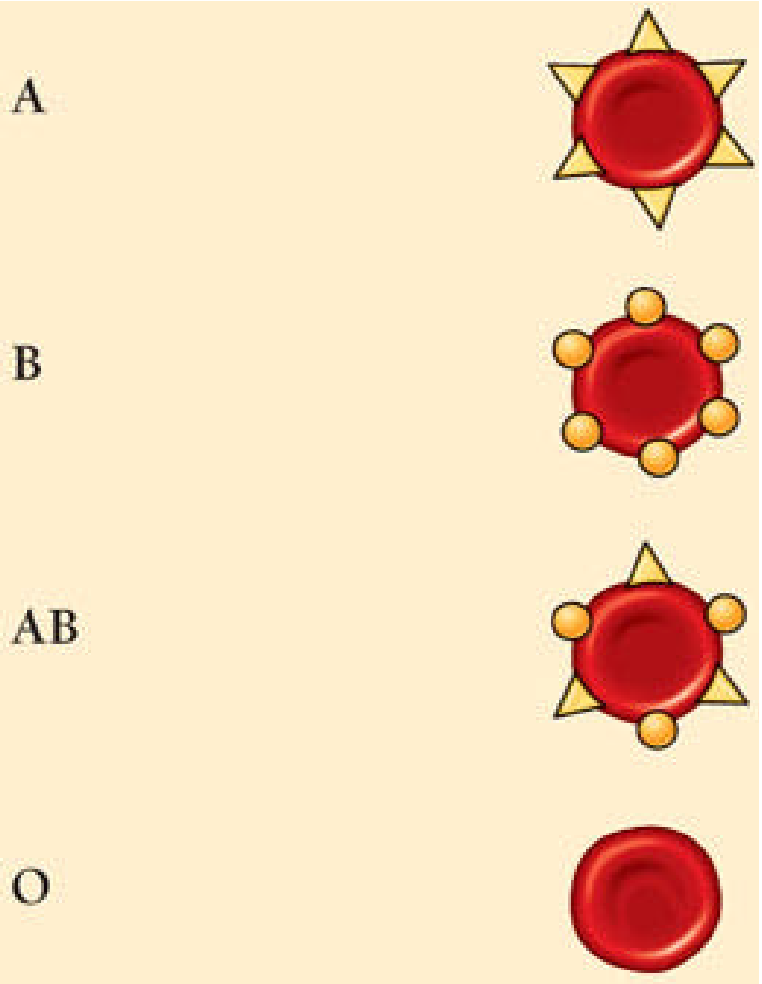


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Genetics of ABO

- There are four blood types in ABO – A, B, AB, O
- A is dominant (Allele is A)
- B is dominant (Allele is B)
- AB are co-dominant (Genotype is AB)
- O is recessive (Allele is O)



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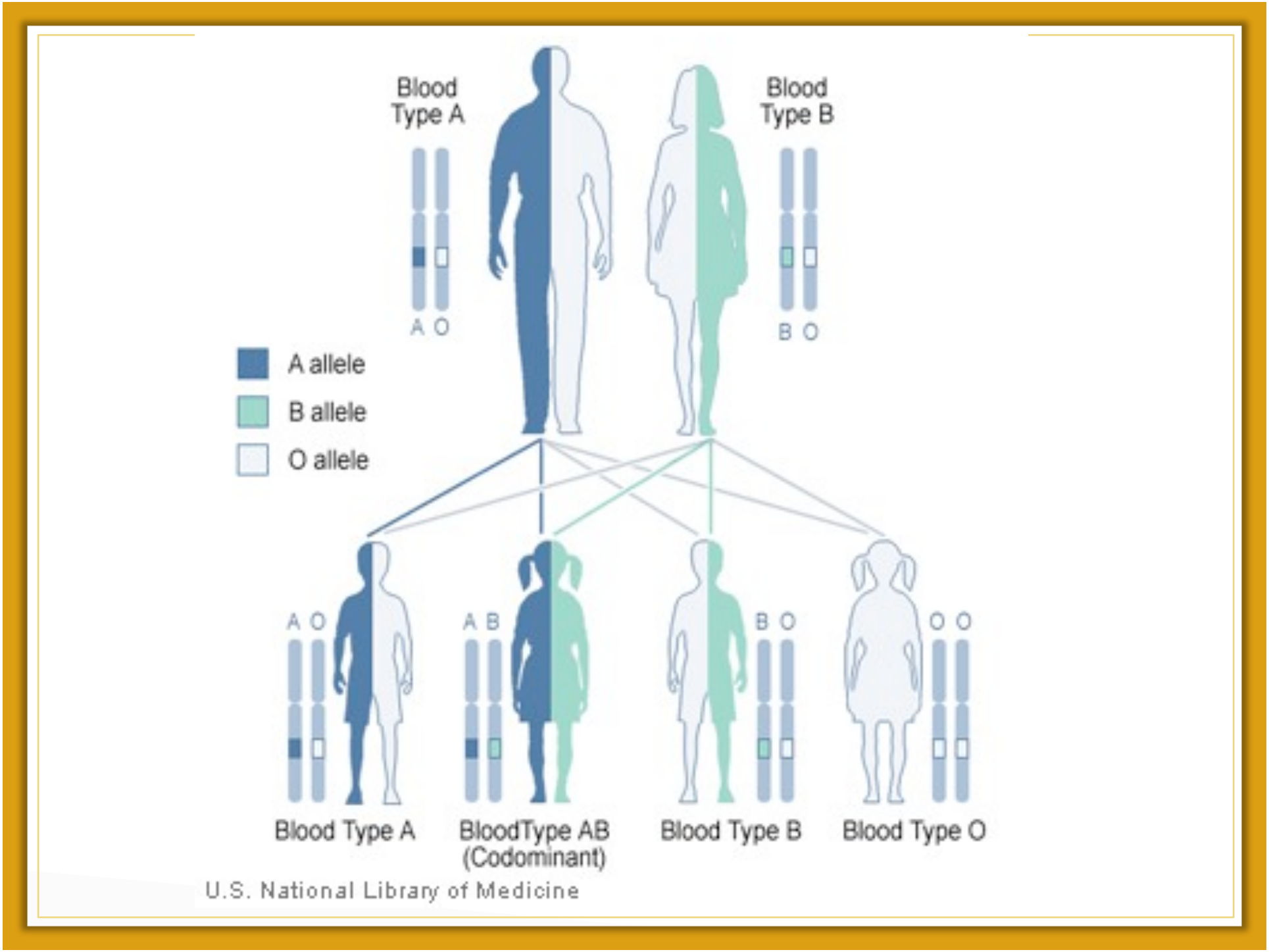
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Inheritance of ABO

Phenotype	Genotype
A	AA or AO
B	BB or BO
AB	AB
O	OO

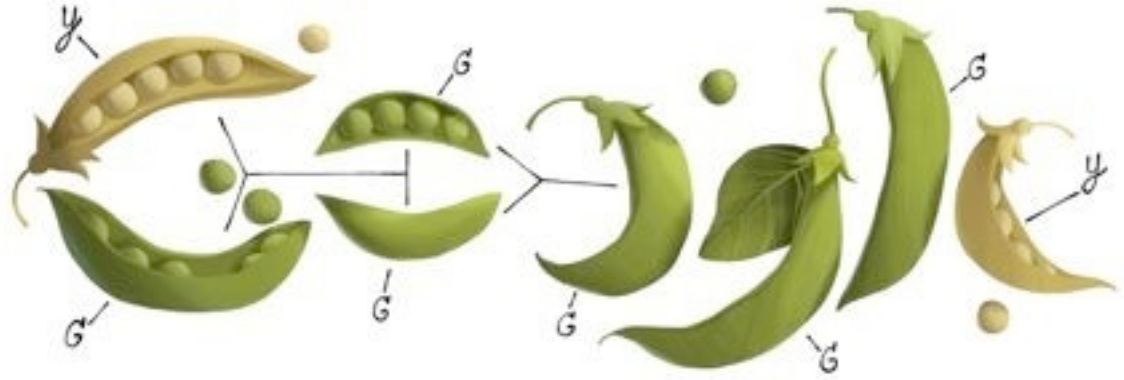
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Google's doodle on Mendel's Birthday

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